

ALA-2**Topic- NESTED IF****1. What is Nested IF?**

- A **nested if** is an if statement inside another if or else block.
- It is used when we need to check multiple conditions step by step.

2. Syntax of Nested IF

if condition1:

if condition2:

 # statements

else:

 # statements

else:

 # statements

3. Why we use Nested IF?

Nested if is used when:

- Multiple conditions depend on each other
- Decision making is complex
- One condition is checked only if another condition is true

4. Check whether a year is a leap year using nested if.

year = 2024

if year % 4 == 0:

 if year % 100 == 0:

 if year % 400 == 0:

 print("Leap Year")

 else:

 print("Not Leap Year")

 else:

```
        print("Leap Year")
else:
    print("Not Leap Year")
```

5. Find the smallest number among three numbers.

```
a = 10
b = 5
c = 8
if a < b:
    if a < c:
        print("A is smallest")
    else:
        print("C is smallest")
else:
    if b < c:
        print("B is smallest")
    else:
        print("C is smallest")
```

6. Check login using username and password.

```
username = "admin"
password = "1234"
if username == "admin":
    if password == "1234":
        print("Login Successful")
    else:
        print("Wrong Password")
else:
    print("Invalid Username")
```

7. Check whether a character is uppercase, lowercase, or digit.

```
ch = 'A'
if ch.isalpha():
    if ch.isupper():
        print("Uppercase")
    else:
        print("Lowercase")
else:
    if ch.isdigit():
        print("Digit")
    else:
        print("Special Character")
```

8. Calculate bill based on units.

```
units = 120
if units <= 100:
    bill = units * 2
else:
    if units <= 200:
        bill = 100 * 2 + (units - 100) * 3
    else:
        bill = 100 * 2 + 100 * 3 + (units - 200) * 5
print("Bill:", bill)
```

9. Check if a person can vote and has valid ID.

```
age = 20
has_id = True
if age >= 18:
    if has_id:
        print("Eligible to Vote")
```

```
else:  
    print("ID Required")  
else:  
    print("Not Eligible")
```

10. Check whether triangle is Equilateral, Isosceles, or Scalene.

```
a = 5  
b = 5  
c = 5  
if a == b:  
    if b == c:  
        print("Equilateral Triangle")  
    else:  
        print("Isosceles Triangle")  
else:  
    if b == c or a == c:  
        print("Isosceles Triangle")  
    else:  
        print("Scalene Triangle")
```

Syntax

◆ Basic Syntax of Nested IF

```
if condition1:  
    if condition2:  
        # statements  
    else:  
        # statements  
else:  
    # statements
```

Extended Syntax (Multiple Nested Levels)

if condition1:

 if condition2:

 if condition3:

 # statements

 else:

 # statements

 else:

 # statements

else:

 # statements